

Assignment 3 - 2018

CS232/NetSys201/EECS248A 2018

Deadline: December 13th 11:59pm

P1 (25 Points) A router implements a preemptive priority queueing policy, where high priority packets are served first, and their arrival interrupts low priority packets' service. If the service is interrupted, the packet is dropped. The service time of packets is an exponential variable with rate $\mu=4$ pkt/s. The input traffic to a router has interarrival time distributed according to an exponential distribution with rate $\lambda_1=2$ pkt/s (high priority) and $\lambda_2=1$ pkt/s (low priority). At time $t=0$, a packet (packet A) is being served by the router. Assume that only one more packet arrives in the router's buffer (packet B), that is, no more packets arrive after this packet. Note that packet A and B can be either high or low priority. a) compute the probability that the service of packet A is interrupted. b) compute the time from the arrival of packet B to the time it leaves the system.

P2 (35 Points) A server accepts at most 3 TCP connections (*i.e.*, at most 3 connections can be active simultaneously). The duration of a connection is a random variable distributed according to an exponential distribution with rate $\mu=4$ connection/min. Connection request interarrivals is a random variable distributed according to an exponential distribution with rate $\lambda=2$ connection/min.

- Compute the probability that an incoming request is not accepted.
- Assume two connections are active, compute the probability that the next request arrives before one of the two connections terminates.
- Assume one connection is active. Find the probability that at least 2 new connection requests arrive before that connection terminates.

P3 (15 Points) Consider a $M/M/1/K$ system with arrival rate $\lambda = 1$ pkt/s and service rate $\mu = 2$ pkt/s.

- Set $K=1$ and compute the probability P_b that a packet will be dropped upon arrival.
- Now set $K=2$ and compute P_b . Is P_b smaller or bigger than in (a)?
- Set $K=1$, and increase μ to 4 pkt/s and compute P_b . Is P_b smaller or bigger than in (a)?

P4 (25 Points) Consider a $M/M/1/\infty$ system, where $\lambda = 1$ pkt/s and $\mu = 2$ pkt/s (that is, the system has an infinite buffer)

- Compute the expected total delay experienced by packets going through the system.
- Now consider N separate $M/M/1/\infty$ systems in parallel, where arrivals are equally distributed among their respective buffers and the service rate of the individual servers is μ/N .
 - Compute the effective arrival rate in each individual buffer.
 - Compute the expected total delay experienced by packets going through the system (from arrival in any queue to departure). Is the delay smaller or bigger compared to (a)?

<https://www.chegg.com/homework-help/questions-and-answers/consider-m-m-1-system-1-pkt-s-2-pkt-s-system-infinite-buffer-compute-expected-total-delay--q33751748?trackid=2a640e5ea9d6&strackid=cb9039a36e50&ii=1>